AKKAS MUGHAL

(226) · 989 · 3130 · akkasmughal94@gmail.com

SKILL SUMMARY

Programming & Scripting	VARIAN, VBA, C#, C++, Python, MATLAB, Java, SQL, Test Scripts
Embedded & Control Systems	DCCs, PLCs (Siemens Step 7), FPGA, ARM, Integrated Circuits
Simulation, Testing & Validation	XVF, IOSIM, DCC-Z, Cadence, Field Testing, Hardware-in-the-Loop
Tools & Processes	Version Control (TFS), MAXIMO, Technical Documentation, Root Cause Analysis, Configuration Management

EDUCATION

University of Waterloo — Sept. 2012 - April 2018
BASc in Electrical Engineering (Co-op)

EXPERIENCE

Bruce Power — Sept. 2018 - Aug. 2025
Computer Design Engineer — *Tiverton, ON*

Design & Troubleshooting

- Designed and implemented hardware and software system modifications supporting multiple plant systems.
- Led complex troubleshooting of mission-critical digital control systems during live plant operations and outages, reducing system downtime and operational risk in a safety-critical environment.
- Validated design changes using non-operational hardware and emulation tools prior to deployment.
- Coordinated system modifications across projects, minimizing configuration conflicts and integration risks.
- Prepared and verified engineering design packages (drawings, OLW, specifications, test plans, and reports) in compliance with regulatory standards.
- Prioritized and resolved software and hardware system defects based on severity and operational impact.

Responsible System Engineer

- Owned system health for assigned digital control systems through daily, quarterly, and annual reviews.
- Produced formal system health reports and sustainment strategies, guiding maintenance and investment decisions.
- Developed action and mitigation plans to improve system reliability and support planned and unplanned outages.
- Planned and scheduled preventative maintenance and deficiency corrections, ensuring operational readiness.

Project & Commissioning Support

- Deployed software modifications across plant systems including DCCs, contact scanners, AR, and PI platforms.
- Supported cold & hot commissioning by troubleshooting field issues and validating requirements.
- Supported system available for field service and turnover activities, enabling safe handover to operations.
- Implemented cybersecurity controls and verified system backups for mission-critical systems.

General Engineering Support

- Developed and maintained digital control system test environments and simulators, improving pre-deployment validation accuracy and reducing production defects.
- Maintained software programs, integrating enhancements and deploying patches across operational systems.
- Generated source, object, executable, and listing files for both offline diagnostics and online DCC deployments.
- Utilized version control (TFS) and asset management systems (MAXIMO) to manage system configurations.
- Ensured alignment between plant hardware/software configurations and engineering documentation.
- Produced calibration documentation for flux detector systems in accordance with F&P and SMP guidelines.
- Mentored junior engineers and delivered technical training materials to diverse groups of personnel.

General Motors — Sept. 2017 - Dec. 2017
Controls Engineer (Co-op) — *St. Catherines, ON*

- Modified PLC ladder logic programs using Siemens Step 7 to support updated manufacturing processes.
- Reworked assembly line operations during planned outages in coordination with maintenance personnel.
- Reduced production idle time by expanding pallet buffer capacity within existing floor space constraints.